

Collected Results, Figures, Tables, and Statistics

When Does Multimodal Diffusion Inverse Kinematics Help Soft Continuum Arms—and Does It Survive Model Mismatch?

Numbering matches the paper exactly. Every value traces to a committed JSON in the repository root.

How the model works, once, up front. The task is inverse kinematics for a four-segment pneumatic continuum arm. Twelve chamber pressures determine a three-dimensional tip position, so many distinct actuations reach the same target and the inverse map is one-to-many. A regressor trained with mean squared error fits the average of those valid answers, and the average of distinct arm shapes is itself an invalid shape. The diffusion model avoids this by never regressing the actuation directly. During training it corrupts each dataset actuation with Gaussian noise at a random noise level and trains a residual network, conditioned on the noise level and on the tip target through feature-wise modulation, to predict the injected noise. Learning to denoise at every noise level is equivalent to learning the gradient field of the full conditional distribution over actuations, which preserves every solution mode instead of their average. At query time the model starts from 32 independent Gaussian samples and deterministically refines each one through 20 denoising steps, so different starting noise flows into different solution modes and drawing samples is itself an enumeration of solutions. Classifier-free guidance, trained by dropping the conditioning ten percent of the time and applied at weight 1.5, sharpens accuracy while retaining diversity. Each figure, table, and statistic below states how this mechanism produces the result and what the result contributes to the paper.

Table I. Core comparison on the PCC arm

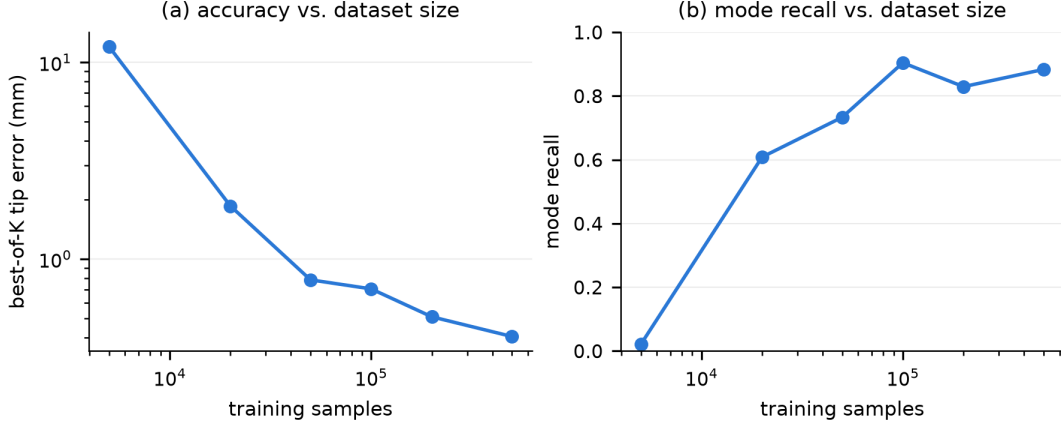
	Diffusion	grad-IK	MLP	MDN
Tip err. best-of- K (mm)	0.44 ± 0.03	0.013	13.4 ± 0.0	17.3 ± 0.2
Tip err. single sample (mm)	1.69 ± 0.15	—	13.4 ± 0.0	83.6 ± 2.7
Success @ 5 mm	1.00	1.00	0.060	0.027
Mode recall	0.84 ± 0.03	0.85	0.00	0.00
Diversity (curv. space)	28.2 ± 0.2	29.0	0.0	0.0
Obstacle success	0.978 ± 0.002	0.980	0.040	0.018
Query time (ms/target)	38.6 ± 4.9	428	0.013	0.022

(mean $\pm$ std over 3 training seeds, $K=32$ samples per target, 512 held-out targets for accuracy, 20 targets for mode recall, 200 obstacle trials. grad-IK is one optimization run per target. A single-target diffusion query costs about 84 ms.)

The diffusion model samples 32 candidate actuations per target and the best one lands within 0.44 mm of the target on average, with every target solved at the 5 mm tolerance. Because the denoiser learned the whole conditional distribution rather than its mean, its samples cover 84 percent of the independently enumerated true solution modes, while the deterministic MLP and the mixture density network recover none. The MLP fails because a single output cannot represent multiple answers, and the MDN fails because none of its mixture components is precise enough to pass the tolerance. The classical optimizer grad-IK, given the true simulator to optimize through at query time, matches or slightly exceeds the diffusion model on every task metric, which shows that the learned model’s real advantage is answering from data alone at 5 to 11 times lower query cost with no simulator in the loop. This table is the paper’s foundation.

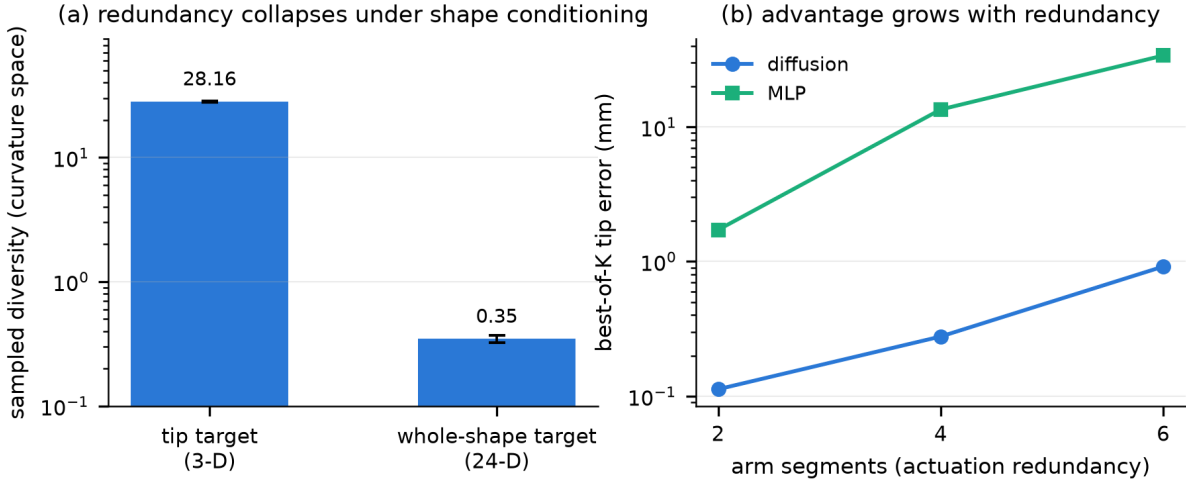
It establishes the multimodality advantage over learned baselines, and it honestly scopes the comparison against classical optimization so that the transfer experiments later in the paper test exactly the property, amortization, that makes the learned model worth having.

Fig. 1. Dataset-size ablation



The same diffusion architecture was retrained from scratch on datasets from 5,000 up to 500,000 motor-babbling samples, single seed. Best-of- K tip error and mode recall both saturate once the dataset reaches roughly 50,000 to 100,000 samples. The denoiser needs enough coverage of the actuation space to learn the score field over all solution branches, and beyond that coverage additional data adds nothing. The significance for the paper is methodological. It shows the headline results are not starved for data, so the baselines' failures cannot be blamed on dataset size, and it tells a practitioner the data budget at which this approach becomes viable.

Fig. 2. The advantage tracks redundancy



Panel (a) shows the shape-conditioning manipulation. Retraining the same models with the 24-dimensional whole-arm shape as the conditioning signal instead of the 3-dimensional tip signal removes almost all redundancy, because specifying the whole shape pins down the actuation. The diffusion model's sampled diversity collapses from 28.16 ± 0.22 to 0.346 ± 0.022 , a factor of about 80, and the MLP goes from about 30 times worse than diffusion to only about 3 times worse, with both models at 100 percent success. Panel (b) shows the morphology manipulation. Training

fresh models on two-segment and six-segment versions of the arm adds or removes actuation freedom, and the accuracy gap widens monotonically with segment count, with the obstacle-task gap reaching 100 percent versus 1 percent at six segments. The mechanism is that the denoiser represents whatever conditional distribution the data presents. When conditioning removes the multimodality, diffusion’s samples concentrate onto the single answer and a deterministic regressor is nearly as good. When morphology adds multimodality, mode averaging hurts the regressor more while diffusion degrades gracefully. These two manipulations are the paper’s causal evidence that the advantage in Table I is specifically redundancy resolution and not generic model capacity.

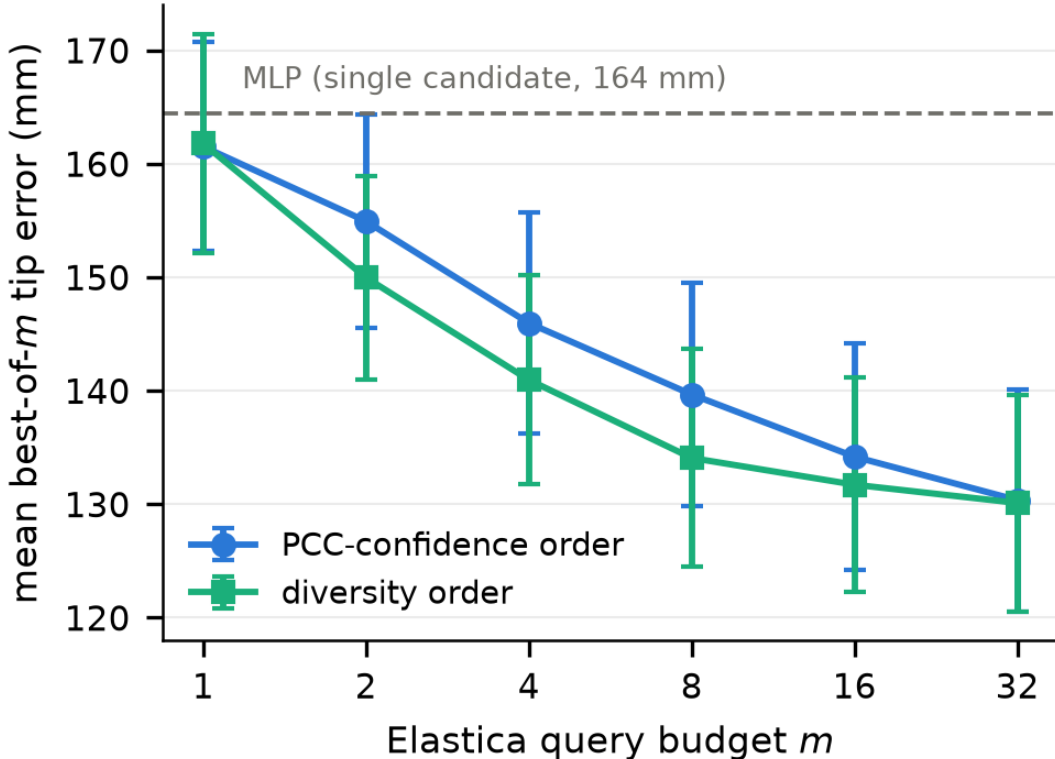
Table II. Recovery mechanisms at a glance (transfer domain)

Mechanism	err (mm)	succ.	Cost / caveat
none (top-1 candidate)	161.5	0%	no target queries
best-of-32 selection	130.4	0%	32 queries/target
+ diversity order	−6.7 at $m=8$	0%	same queries, $p \approx 0.002$
guided sampling	129.0	0%	400 labels, suggestive
gain randomization	140.6	0%	degrades source, no gain
fine-tune, 0% mix	67.1 ± 0.9	10%	forgets source (133 mm)
fine-tune, $\geq 5\%$ mix	≥ 133.8	$\leq 10\%$	retains source, forfeits gain

(each row uses its own study’s held-out target set of 15 to 45 targets. The 139.9 mm collapse figure quoted in the text corresponds to best-of-8 selection.)

The transfer experiment executes actuations sampled from the PCC-trained diffusion model on a Cosserat-rod simulation of the same arm, where accuracy collapses from 0.67 ± 0.03 mm to 139.9 ± 1.8 mm with zero success on every seed. This table collects every mechanism the paper tests for recovering that loss. Selection exploits the one thing a multimodal model uniquely offers, many candidates to test in the target domain, but even 32 target-domain evaluations per target leave the error at 26 times the tolerance. Steering generation with a learned transfer-error regressor gives a small held-out gain that does not survive multiple-comparison correction. Randomizing the curvature gain during training recovers exactly nothing because the mismatch between the simulators is directional rather than a matter of scale. Only fine-tuning on target-domain data produces nonzero success, and only in its most destructive form. The significance is the paper’s second headline. The learned inverse map is simulator specific, no inference-time mechanism comes close to fixing it, and the one training-time mechanism that works destroys the source domain, which motivates parameter-level retention methods as the necessary next step.

Fig. 3. Candidate selection under a query budget



For 45 held-out targets the model samples 32 candidates each, and a budget of m expensive target-domain evaluations selects the best candidate. Mean error falls from 161.5 mm at $m=1$ to 130.4 mm at $m=32$, improving all 45 of 45 targets, yet success stays at zero at every budget. The candidates are diverse because the denoiser spreads its samples across the source domain’s solution modes, and some of that spread happens to land closer to the target domain’s very different solution manifold, which is why selection helps broadly but shallowly. A greedy farthest-point ordering of the budget in curvature space beats spending the budget in confidence order by 6.7 mm at $m=8$, the paper’s strongest statistical effect. The significance is a precise characterization of what multimodality buys under model mismatch. It buys a real, broad, but bounded improvement, and how the budget is spent matters more than how large it is, while the transferred task itself remains unsolved.

Table III. Fine-tuning and the retention–transfer cliff

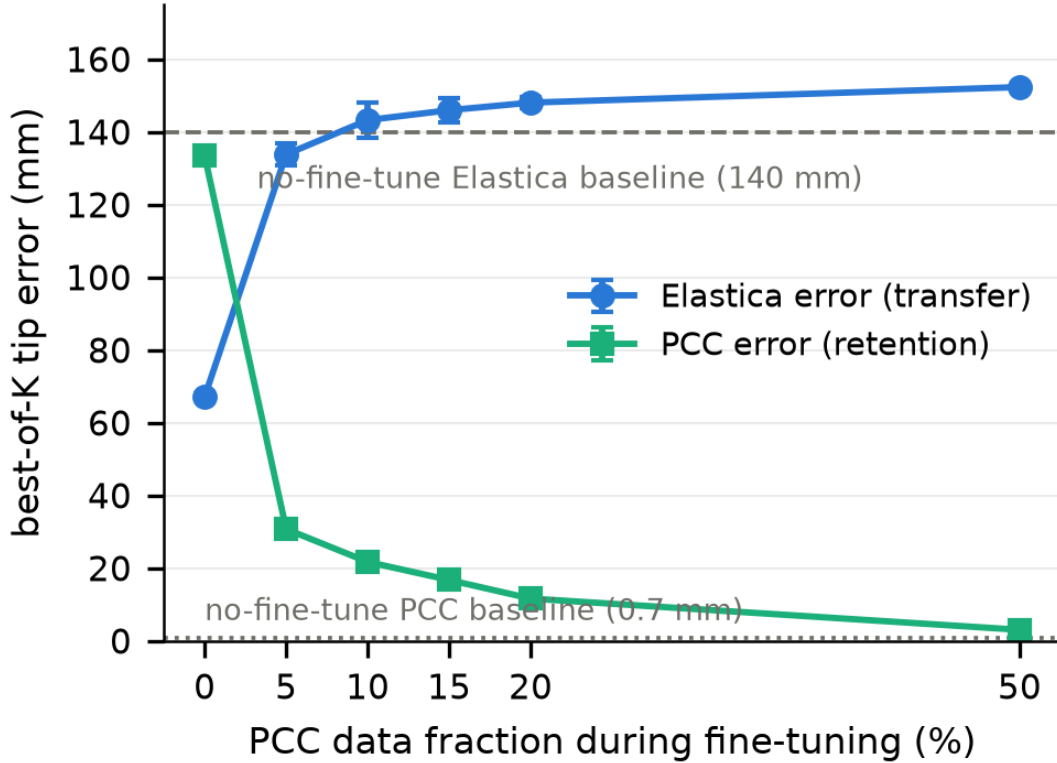
PCC mix	PCC err (mm)	succ.	Elastica err (mm)	succ.
none (baseline)	0.44 ± 0.03	100%	151.1 ± 2.5	0%
0%	133.4 ± 1.2	0%	67.1 ± 0.9	10%
5%	30.7 ± 0.6	6%	133.8 ± 3.1	8%
10%	21.7 ± 0.5	17%	143.2 ± 4.9	8%
15%	16.7 ± 0.2	26%	146.0 ± 3.4	8%
20%	11.6 ± 0.3	51%	148.0 ± 1.6	10%
50%	3.0 ± 0.4	96%	152.3 ± 0.4	3%

(mean $\pm$ std, 3 seeds. PCC columns use the 512-target evaluation, Elastica columns use the same 20

held-out targets for every row including the baseline.)

The pretrained diffusion model is fine-tuned on 10,000 Elastica-simulated samples for 20 epochs at learning rate 3×10^{-5} , with the stated fraction of each batch drawn from the original PCC data through weighted sampling. Fine-tuning purely on target data rewrites the denoiser’s score field toward the target domain’s solution distribution, which produces the only nonzero transfer success in the paper, 67.1 mm and 10 percent, at the price of catastrophic forgetting of the source domain. Mixing source data back in restores source accuracy smoothly and monotonically down the left columns, but the transfer gain does not trade off smoothly. At just 5 percent source data the transfer error has already rebounded to 133.8 mm, forfeiting about 80 percent of the improvement over the matched-target baseline of 151.1 mm, and it drifts only slowly after that. The significance is that retention and transfer sit on a cliff rather than a tradeoff curve, so no data-mixing ratio is a sweet spot, and the paper’s closing argument follows directly. Reshaping the model with small amounts of target data requires protecting parameters, not rebalancing batches.

Fig. 4. The retention–transfer cliff



This figure plots the two right columns of Table III against the mixing ratio with three-seed error bars. The source-domain curve improves smoothly and monotonically as source data is mixed back in, while the target-domain curve loses nearly the entire fine-tuning gain by the first grid point at 5 percent and stays flat afterward. The seed-to-seed standard deviations of at most 4.9 mm show the cliff is not an unlucky training run. Its significance is visual proof of the paper’s final claim, the absence of any data-mixing sweet spot, located by a sweep fine enough to place the cliff’s edge below 5 percent.

All remaining statistics, by paper section

Section III, Setup

- Ground-truth solution sets average 2.05 modes per target, range 0 to 6, from a 2,000,000-sample sweep clustered with DBSCAN in curvature space ($\epsilon=12$, `min_samples` = 5, match radius 15).
- Settle-time convergence check, 8 random actuations. Tip shifts of 50 to 145 mm persist between integration horizons of 1.5, 3, 6, and 9 seconds.
- Couple-gain calibration sweep over 5×10^{-4} to 5×10^{-2} . Mean tip discrepancy between simulators never falls below about 150 mm while workspace radii stay similar (0.20 m versus 0.10 to 0.24 m).

These numbers validate the evaluation machinery itself. Clustering in curvature space matters because pressures contain a per-segment null space that would corrupt distances. The settle check shows the target domain is a fixed deterministic dynamic snapshot rather than an equilibrium, applied identically everywhere, and the gain sweep shows the domain gap is directional rather than a fixable unit mismatch. Together they certify that the transfer collapse measured later is a property of the learned map, not an artifact of calibration.

Section IV, core experiments and ablations

- Seed separation. On every task metric the worst diffusion seed beats the best learned-baseline seed, for tip error 0.49 versus 13.33 mm, a gap exceeding 400 times the across-seed standard deviation.
- Guidance-weight ablation. Error 49.6 mm at weight 0, best 0.22 mm at 1.5, degrading to 3.1 mm at 3 and 13.2 mm at 4 with diversity collapsing from 28.7 to 15.7. An inverted U with the operating point at its peak.
- DDIM steps. Accuracy and recall saturate by about 20 steps, a 3 times speedup over 50 steps at no measurable cost.
- MDN fairness sweep over 4, 8, 16, 32 components. Best-of- K error improves only from 18.7 to 14.8 mm, never surpassing the plain MLP, and mode recall is exactly zero at every setting.
- MDN pathology. Training NLL decreases monotonically at every epoch, no variance reaches its clamp floor, 7.8 of 8 components remain effective, yet the best component mean per target carries 7.5 mm error against the 5 mm tolerance, median component 9.2 mm.
- Mode-enumeration sensitivity, 3 by 3 grid over ϵ in {10, 12, 14} and match radius in {10, 15, 20}. Learned baselines recall zero at all nine settings. Diffusion is higher at three settings, grad-IK at five, one tie.
- grad-IK compute ablation. At 100 iterations and 32 restarts it reaches 0.30 mm at 100 percent success in about 130 ms per target. Restart-RNG variance across three seeds has standard deviation below 0.005 mm.

These statistics defend Table I against every fairness objection the authors could construct. The seed-separation figure replaces formal testing where three seeds would make it hollow. The two ablations show the operating point was chosen at a validated optimum rather than tuned against the baselines. The MDN analyses prove the multimodal baseline fails for a substantive reason, imprecision, rather than a training pathology or an under-tuned component count. The

sensitivity grid shows the central qualitative conclusion, multimodal methods recover modes and learned regressors recover none, at every reasonable setting of the enumeration hyperparameters. The grad-IK numbers quantify the honest cost comparison that scopes the paper’s claims.

Section V, mechanism manipulations

- Shape conditioning, 3 seeds. Diversity 28.16 ± 0.22 to 0.346 ± 0.022 , about 80 times. Accuracy gap shrinks from about 30 times (13.4 versus 0.44 mm tip error) to about 3 times (0.49 versus 0.16 mm mean per-backbone-point error), both models at 100 percent success.
- Morphology, single seed per arm. MLP error 1.7 mm at 75 percent success with two segments, 13.5 mm at 5 percent with four, 33.9 mm at 1.2 percent with six. Diffusion 0.11, 0.28, 0.92 mm with at least 99.8 percent success throughout. Obstacle task at six segments, 100 percent versus 1 percent.

These are the quantitative backbone of Fig. 2, showing both directions of the causal manipulation with the numbers the figure plots. Their significance is that the paper’s central advantage rises and falls with the amount of actuation redundancy and with nothing else that changed in the experiments.

Section VI, transfer and recovery

- Transfer collapse, 3 seeds, 20 targets, $K=8$. Diffusion 0.67 ± 0.03 mm at 100 percent on PCC falls to 139.9 ± 1.8 mm at 0 percent on Elastica. The MLP falls from 13.8 to 164.4 mm.
- Query budget, $n=45$. Error 161.5 ($m=1$), 154.9, 146.0, 139.7, 134.2, 130.4 mm ($m=32$). All 45 targets improve. Final gap to tolerance is about 26 times.
- Ordering comparison, paired within candidate sets. Diversity order beats confidence order by 6.7 mm at $m=8$ ($t=3.34$, $p \approx 0.002$, 24 targets better versus 12) and 3.3 mm at $m=16$ ($t=3.21$).
- Predictor correlations at $n=45$, sized to detect $r \geq 0.41$ at 80 percent power. Candidate diversity versus improvement $r=0.20$ ($p=0.19$), Jacobian minimum singular value versus improvement $r=0.02$ ($p=0.88$). A pilot effect at $n=20$ did not replicate.
- Guided sampling, 15 held-out targets. Best-of-16 error 134.6 to 129.0 mm, paired improvement 5.6 ± 2.5 mm ($t=2.27$, $p \approx 0.04$, 11 of 15 targets). Of the roughly six hypothesis tests in the paper only the ordering result survives Bonferroni correction.
- Combined 2 by 2 factorial, guidance by ordering. The mechanisms do not stack. Guidance concentrates the candidate distribution while diversity-ordering spreads selection.
- Domain randomization, gain drawn uniformly from 15 to 35 per sample. PCC accuracy degrades 0.44 to 2.18 mm and transfer is unchanged, 140.6 to 140.6 mm.
- Fine-tune baseline matched to the fine-tuning target set. The base models score 151.1 ± 2.5 mm at 0 percent on those 20 targets, making the best fine-tuned result a 56 percent error reduction, and the 5 percent mix forfeits about 80 percent of it.

This block is the complete statistical record of the paper’s second arc. The collapse numbers establish the problem on every seed. The budget and ordering numbers quantify the ceiling of selection and identify how to spend queries, with the ordering effect standing as the paper’s only correction-surviving significance. The null correlations, reported with their power analysis, honestly retire a pilot hypothesis. The guidance, factorial, and randomization results bound

what steering and synthetic variation can do. The matched baseline puts the fine-tuning gain and the cliff on the same footing, closing the loop on the paper’s final claim that data mixing cannot smooth the retention-transfer tension.

Efficiency summary

- Diffusion 38.6 ms per target with 64 targets batched, about 84 ms for a single-target query, 20 DDIM steps, CPU.
- grad-IK 428 ms per target at the full budget, about 130 ms at the reduced budget that still reaches 0.30 mm.
- MLP 0.013 ms and MDN 0.022 ms per target, fastest by far and unable to perform the task.

The timing numbers scope the amortization claim honestly, a 5 to 11 times advantage over the classical optimizer depending on batching, and they matter because amortization without simulator access is precisely the property the transfer experiments then show does not survive a change of dynamics model.